

1.4.7. No more HAL complications

Under Windows XP, different hardware abstraction layers called for multiple images to configure and deploy. You might have thus needed two or more images for a single platform; each additional image bringing added costs and complexity.

Windows Vista can now detect which HAL is required and can automatically install it—you are no longer forced to create iterations for different HALs.

1.4.8. Windows PE to rule them all

For a while now, Windows PE (Pre-installation Environment) has been the deployment tool of choice for various organisations. However, you need to be a Software Assurance partner with Microsoft in order to get access to this tool. This is no longer the case under Vista: you can now download Windows PE 2.0 from Microsoft's Web site and use it freely to deploy licensed copies of Vista. Windows PE 2.0 offers better performance from 32-bit and 64-bit networking stacks and tools. It also supports tools such as Windows Scripting Host and VBScript.

Much like Vista, Windows PE 2.0 comes as an image that is componentised, and can be serviced both online and offline. While it offers some new components such as language packs, font packs, Windows Recovery Environment, and MSXML 3.0, other components can be easily added using available tools `peimg.exe`—including mass storage devices, which no longer require any special handling.

1.4.9 The WIM format

The image format used under Vista is file-based, and is called the Windows Imaging (WIM) format. Being file-based, the images can be applied to a partition non-destructively. Thus during deployment, the user state can be saved locally instead of on a network server—which will reduce network traffic while deploying the OS. During deployment you may create an image file that contains the contents of either a single disk volume or a partition. Thus to capture a sys-